Zoo101 paper,,,
Normality
Refugal force
PCR
Traditional flow and acoustic flow
Dry preservation
Function of blood cell
Specimen Wala b long

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8:59 AM

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Sorry centrifugal force and angular velocity

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Zoo101 paper,,,
Normality
Refugal force
PCR
Traditional flow and acoustic flow
Dry preservation
Function of blood cell
Specimen Wala b long

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Sorry centrifugal force and angular velocity

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To day may pppr zoo 101
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Zoo101
Mcqs mostly from gel electroporocis . dry preservation. transition, preservation, DNA probe, traditional and acoustic flow,function of blood. Detection of genome. Etc

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To day my ppr

Zoo 301 ....,,,,,,,,,
Passive immunity ,,,
Transport system of vertebrate,,,,,,,
Function of kidney,,,,
What type of respiratory organ in arthopoads and malloscs ,,,,,
What are arteries ,,,,
What is second line of defence ,,,,,,,

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What is antigens. What is diaphragm. What is serum. What is tympanum lucenuied what is chemical messenger.

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Zoo101 today paper
Short
Functions of types of blood cell
Normality
Dry preservation
PCR
Multiplex PCR
Long
Specimen wala
Centrifugal force and angular velocity

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Define length? How many steps in concentration? How many functions of blood cell? Mcq how many types of centrifuges , North blot method kes ki sequence detect kerta hua ya Mary zoo 101 ka paper ma aya tha bake questions yad nahi

12:55 PM

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40 mcq thy or bake questions thy

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Total 60 mark ka tha 40 mcq
20 mark ka object tha 2 , 3, 5
Marks ka questions thay I think
ziada paper lecture no 45 say 90
lecture may say aya tha Sanger
sequence, concentration step ,
preservation of sample function
of blood, Fahrenheit ko Celsius
ma change kerna,max am Gilbert,
ya question Mary paper ma thay

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Blood cell type function ,pcr define, dry preservation, spectrophotometer , diameter and it's step, explain transition type.

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Preservation of specimen

7:05 PM

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define probe of dna

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A **probe** is a single-stranded sequence of **DNA** or RNA used to search for its complementary sequence in a sample genome. The **probe** is placed into contact with the sample under conditions that allow the **probe** sequence to hybridize with its complementary sequence.

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dna molalty



People also ask

What is the molar concentration of DNA?

Hence, the **molar concentration of DNA** in each diploid cell in humans is 2.77×1023moles.

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dna molalty

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Showing results for dna ***molarity***

Search instead for dna molalty

3. Calculate the **molar concentration** of DNA using the following equation: Picomoles/µl = DNA **Concentration**(µg/ml) / [0.66*DNA Size(bp)]

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Nothern Blotting

A northern blot is a laboratory method used to detect specific RNA molecules among a mixture of RNA. Northern blotting can be used to analyze a sample of RNA from a particular tissue or cell type in order to measure the RNA expression of particular genes.

The first step in a northern blot is to denature, or separate, the RNA within the sample into single strands, which ensures that the strands are unfolded and that there is no bonding between strands. The RNA molecules are then separated according to their sizes using a method called gel electrophoresis. Following separation, the RNA is transferred from the gel onto a blotting membrane. (Although this step is what gives the technique the name "northern blotting," the term is typically used to describe the entire procedure.)

Once the transfer is complete, the blotting membrane carries all of the RNA bands originally on the gel. Next, the membrane is treated with a small piece of DNA or RNA called a probe, which has been designed to have a sequence that is complementary to a particular RNA sequence in the sample; this allows the probe to hybridize, or bind, to a specific RNA fragment on the membrane. In addition, the probe has a label, which is typically a radioactive atom or a fluorescent dye. Thus, following hybridization, the probe permits the RNA molecule of interest to be detected from among the many different RNA molecules on the membrane.

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LQ: What is DNA Sequencing(Maxam Gilbert)

Method

- Maxam-Gilbert method of DNA Sequencing
- It is a method by which the sequence of a DNA fragment is identified by using chemicals, thats cut DNA at specific points.
- Also called Chemical degradation method of DNA sequencing.

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Procedure

it is a fragment of double stranded DNA, and you do not its sequence:

Step :1

- As the sequence of both strands are unknown, but if we find out the sequence of one strand, we would get to know sequence of other one also. At first, the double stranded

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fragment is separeted into two single strands by applying high Temperature or high PH.

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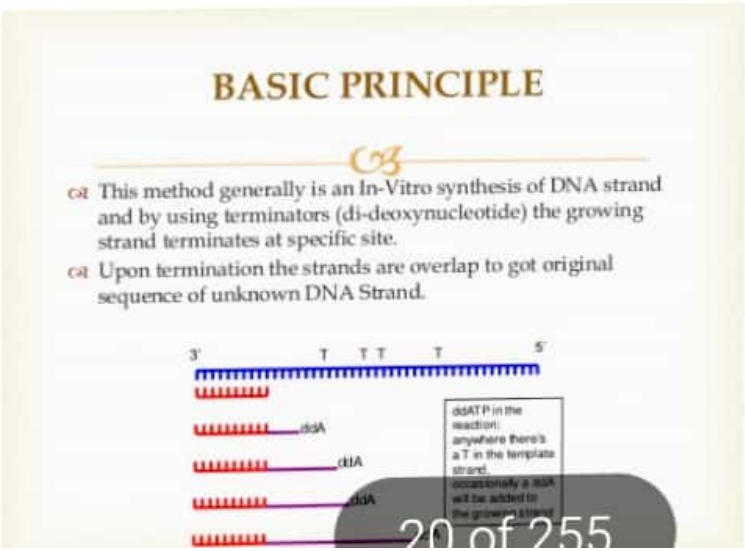
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SQ: What is theb PRINCIPLE of DNA sequencing (Sanger)?

- The 5′ carbon of an "incoming" deoxynucleotide (dNTP) is joined to the 3′ carbon at the end of the chain. Hydroxyl groups in each position form ester linkages with a central phosphate. , the nucleotide chain elongates.
- The key to Sanger's sequencing method is the peculiar chemistry of dideoxynucleotides (ddNTP). Like a deoxynucleotide, a ddNTP is incorporated into a chain by forming a phosphodiester linkage at its 5′ end.
- However, the ddNTP lacks a 3′ hydroxyl group (OH) necessary to form the linkage with an incoming nucleotide.

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inab Arshad



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Lesson 99

Preservation Techniques

SQ: LQ: Write a note on Preservation Techniques And describe the process?

Steps for the preservation of specimens for scientific study are as follows:

- **Euthanizing**. Specimens should be euthanized in a way that will leave them undamaged and relaxed.
- **Injection and slitting**. Liquid preservatives must be introduced into the body cavity, limbs and tail, either by hypodermic injection or through slits.
- **Fixing.** While the specimens are still relaxed, they should be arranged in trays so that they will harden in the proper position.

Labeling. Each specimen should be accompanied by certain data, either attached directly or entered in a notebook with a number corresponding to a numbered tag tied to the specimen.

Storage. After specimens have been fixed in the proper position, they should be stored in liquid preservative for at least several days, after which they may be allowed to remain in the liquid, or transferred to plastic bags for temporary storage

Preserving Solutions

Formalin: If at all possible, formalin should be used for injecting and fixing specimens. Formalin is the commercial name of a solution of formaldehyde gas (CH20) in water.

Alcohol: It is usually sold at a strength of 95% (190 proof).

Preparation:

- If specimens are to be made permanently immune to decomposition, it is necessary that liquid preservative be introduced into the body cavity, limbs and tail within as short a time as possible after the animals have been killed.

This may be accomplished either by injection (with a hypodermic syringe) or by making deep cuts with a sharp scalpel, razor blade or scissors. The

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Frogs and Toads:

- Injection should be made through the belly, directly into the body cavity. If the body is puffed with air, it should be deflated by gently squeezing with the fingers. Very small frogs require only a few drops of preservative; frogs two or three inches long only a few cc. Introduce only enough preservative required to make the specimen look natural--it should not look bloated. It is not necessary to inject the legs of any but the largest frogs. If equipment for injection is not available, a single slit may be made in the abdomen, to one

side of the midline. The slit should be deep enough to allow free access of the preservative into the body cavity



Salamanders. Most salamanders do not require injection or slitting. If your specimens look "caved in" a small amount of preservative may be injected into the body cavity, or a single slit made in the abdomen to permit preservative.

Tadpoles.  Tadpoles and small salamander larvae

should always be preserved in 10% formalin, never in alcohol. Simply drop the tadpoles into formalin while they are still alive. Be sure there is enough preservative to cover them and avoid overcrowding. After 24 hours all the liquid should be drained off and replaced with fresh formalin

Lizards

- Injection should be made through the belly directly into the body cavity. Care should be taken not to use too much, or the body will become unnaturally distended. A series of slits should be made in the under side of the tail with a sharp scalpel or razor blade. The slits should be from 1/8 to 1/4 inch long and about 1/4 inch apart, and should extend from the base of the tail to the tip. Very large lizards must be injected or slit in the thicker portions of the limbs and neck. If space does not permit preservation of very

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how many types of centrifuge

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1. **Small Benchtop**
   - with or without refrigeration
   - slow speed (eg up to 4000 RPM)
   - common in clinical lab (blood/plasma/serum separation)
   - can take approx (up to) 100 tubes, depending on diameter



Types of Centrifuge depends on :
- Maximum speed of sedimentation
- Presence /absence of vacuum
- Temperature control refrigeration.
- Volume of sample and capacity of centrifugation tubes



two types

Generally, there are two **types of centrifuges**: the filtration and sedimentation **centrifuges**.

en.m.wikipedia.org › wiki › Centrif...

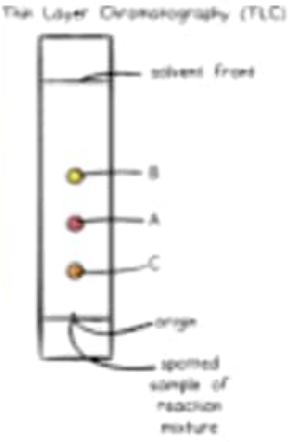
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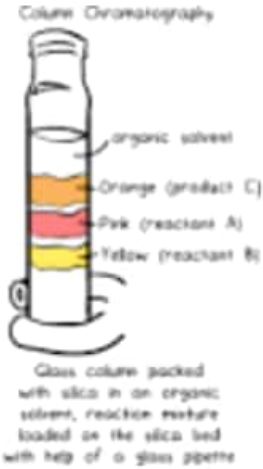
chromatography

Chromatography

Overview | Examples | Videos







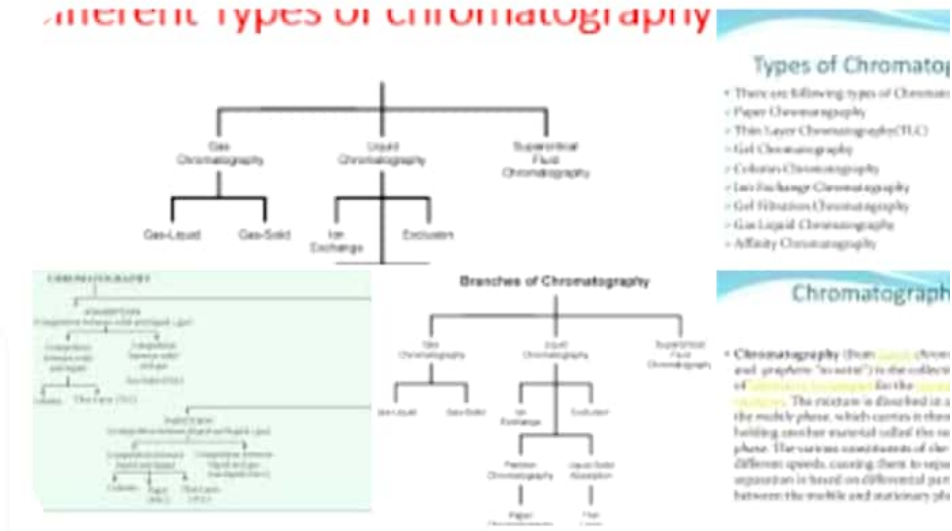


Chromatography is a laboratory technique for the separation of a mixture. The mixture is dissolved in a fluid called the mobile phase, which carries it through a system on which is fixed a material called the stationary phase. Wikipedia

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chromatography types

Types of chromatography



There are four main **types** of **chromatography**. These are Liquid **Chromatography**, Gas **Chromatography**, Thin-Layer **Chromatography** and Paper **Chromatography**. Liquid **Chromatography** is used in the world to test water samples to look for pollution in lakes and rivers.

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ion chromatography

08-Nov-2016

23-Sep-2010

People also ask

How does ion chromatography work?

Ion Chromatography is a method for separating **ions** based upon their interactions with resin (stationary phase) and the eluent (mobile phase). These phases differ between an anion column, which attracts anions, and a cation column, which attracts cations.

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SQ:lq: how many Types of Flow Cytometry ?

1. Traditional flow cytometers

The traditional cytometers are the common cytometer using sheath fluid for focusing the sample stream.

Mcqs: The most common lasers used in traditional flow cytometers are **488 nm (blue), 405nm (violet), 532nm (green), 552nm (green), 561 nm (green-yellow), 640 nm (red) and 355 nm**

(ultraviolet).

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Sq:lq: describe the Types of Flow Cytometry?

2. Acoustic Focusing Cytometers

In these cytometers, ultrasonic waves are used to focus the cells for analysis.

This prevents sample clogging and also allows higher sample inputs.

3. Cell sorters

Cell sorters are a category of traditional flow cytometers which allows the user to collect samples after processing.

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The cells that are positive for the desired parameter can be separated from those that are negative for the parameters.

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Lesson 99

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Alcohol: It is usually sold at a strength of 95% (190 proof).

Preparation:

- If specimens are to be made permanently immune to decomposition, it is necessary that liquid preservative be introduced into the body cavity, limbs and tail within as short a time as possible after the animals have been killed.

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200 of 255

This may be accomplished either by injection (with a hypodermic syringe) or by making deep cuts with a sharp scalpel, razor blade or scissors. The most satisfactory way is by injection. A ten or twenty cc. syringe with a

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Preservation of dry specimen

SQ:LQ: Write a note on Dry preservation

Killing and handling prior to dry mounting

- Insects that are intended to be pinned and stored dry are best killed either in a killing bottle or tube containing a volatile poison, or in a freezer. Freezing avoids the use of chemical killing agents but it is important to place the insects into a small, airtight container to prevent drying out and to freeze them for at least **12–24 h.** Frozen insects must be handled carefully and properly thawed before being pinned, otherwise the brittle appendages may break off. The safest and most readily available liquid killing agent is ethyl acetate, which although flammable, is not especially dangerous unless directly inhaled. It should not be used in an enclosed room. More poisonous substances, such as cyanide and chloroform, should be avoided by all except the most experienced entomologists.
- Ethyl acetate killing containers are made by pouring a thick mixture of plaster of Paris and water into the bottom of a tube or wide-mouthed bottle or jar to a depth of **15–20 mm;** the plaster must be completely dried before use. To "charge" a killing bottle, a small amount of ethyl acetate is poured onto and absorbed by the plaster, which can then be covered with tissue or cellulose wadding. With frequent use, particularly in hot weather, the container will need to be recharged regularly by adding more ethyl acetate. Crumpled tissue placed in the container will prevent insects from contacting and damaging each other. Killing bottles should be kept clean and dry, and insects should be removed as soon as they die to avoid color loss. Moths and butterflies should be killed separately to avoid them contaminating other insects with their scales.

SQ: What is *rigor mortis?*

- Dead insects exhibit *rigor mortis* (stiffening of the muscles), which makes their appendages difficult to handle, and it is usually better to keep them in the killing bottle or in a hydrated atmosphere for 8–24 h (depending on size and species) until they have relaxed, rather than pin them immediately after death. It should be noted that some large insects, especially weevils, may take many hours to die in ethyl acetate vapors and a few insects do not freeze easily and thus may not be killed quickly in a normal household freezer.
- It is important to eviscerate (remove the gut and other internal organs of) large insects or gravid females (especially cockroaches, grasshoppers, katydids, mantids, stick-insects, and very large moths), otherwise the abdomens may rot and the surface of the specimens go greasy. Evisceration,

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How the Preservation of dry specimen can be done ?

- *Spreading and setting* • It is important to display the wings, legs, and antennae of many insects during mounting because features used for identification are often on the appendages. Specimens with open wings and neatly arranged legs and antennae also are more attractive in a collection. Spreading involves holding the appendages away from the body while the specimens are drying.
- Legs and antennae can be held in **semi-natural positions** with pins and the wings can be opened and held out horizontally on a setting board using pieces of tracing paper, cellophane, greaseproof paper, etc.

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- **Setting boards** can be constructed from pieces of polyethylene foam or soft cork glued to sheets of plywood or masonite; several boards with a range of groove and board widths are needed to hold insects of different body sizes and wingspans. Insects must be left to dry thoroughly before removing the pins and/or setting paper, but it is essential to keep the collection data associated correctly with each specimen during drying. A permanent data label must be placed on each macropin below the mounted insect (or its point or stage) after the specimen is removed from the drying or setting

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- board. Sometimes two labels are used — an upper one for the collection data and a second, lower label for the taxonomic identification.

SQ: What is Micropinning (staging or double mounting)

- This is used for many small insects and involves pinning the insect with a micropin to a stage that is mounted on a macropin; micropins are very fine, headless, stainless steel pins 0.5 mm to 1.5 mm long, and stages are small square or rectangular strips of white polyporus pith or synthetic equivalent.

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